IERG4320/IEMS5726: Data Science in Practice

Course Outline

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People

Lecturer Danny Yip @ SHB735 Office Hour: On appointment

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For lecture content, materials, clarifications of course project, reference materials, …

Tutor 1 YANG Bufang @ SHB702

Email: yb022@ie.cuhk.edu.hk

Contact Bufang if you need specific help when working on your assignment 2 and 3

Tutor 2 WANG, Genglin @ SHB 828

Email: wg024@ie.cuhk.edu.hk

Contact Genglin if you need specific help when working on your assignment 1 and 4

Lectures

- Main Lecture
  - Wednesday 7:00 - 9:00 pm @ LSB LT6
- Dates (12 Lectures)
  - 7/1, 14/1, 21/1, 28/1, 4/2, 11/2, 25/2, 4/3, 11/3, 18/3, 25/3, 1/4
- No tutorial components in master course
- Final Examination will be week 15
  - Location: ERB 1004 (IE lab)
  - Date: 15 Apr (Wed)
  - Time: Session A1: 6:00 - 8:00 pm, Session A2: 8:30 - 10:30 pm
  - Open book, open note, close internet, close discussion
- Notes
  - No classes on 18 Feb due to Chinese New Year Holiday

Course Materials

- All course materials are uploaded onto the Blackboard
- http://blackboard.cuhk.edu.hk/
  - General information
  - Announcement
  - Lecture notes and recordings
  - Assignments and course project
- Suggested solutions will not be available
- Please submit all your coursework to the Blackboard. We do not accept coursework via email attachments.

Course Schedule - Lecture

#	Topic	Date
1	Course Outline, FastAPI App	7/1
2	Basic Statistics	14/1
3-5	Data Representation	14, 21, 28/1
6-7	Data Modeling	4, 11/2
8	Data Visualization	25/2

Course Schedule - Lecture

#	Topic	Date
9	Case Study I: Business	4, 11/3
10	Case Study II: Text mining	11, 18/3
11	Case Study III: Multimedia	18, 25/3
12	Case Study IV: Game Applications	25/3, 1/4
13	Course Summary	1/4
#	Final Examination	15/4

Course Assessment

1. **UReply attendance (5%)**
   - Ureply question(s) available during second break (MSc session)
   - Participate one UReply activity = 0.5%
   - No matter your answer is correct or not
   - Remember to backup the questions on your own
2. **4 Assignments (35%)**
   - Both written and programming practices
   - We use Python programming language for the programming exercises
   - Deadlines: 22 Jan (Thur), 5 Feb (Thur), 26 Feb (Thur), 12 Mar (Thur)
3. **Course project (20%)** due on 8 May (Fri)
4. **Final examination (40%)**
   - Written exam, open books/notes, close internet/discussion
   - Date: 15 Apr (Wed), Location: ERB 1004
   - Session A1: 6:00 - 8:00 pm, Session A2: 8:30 - 10:30 pm

More about the course project

- Decide your own data science course project
- Understand the steps in data science pipeline
- Deploy a service using both Python programming and deep learning

You will fail the course if …

1. Your final examination performance is unsatisfactory (e.g. below 30%)
2. Your grand total cannot meet the passing line (e.g. below 40%)
3. You do not submit the course project

No cheating, no plagiarisms

University Guidelines to Academic Honesty

- http://www.cuhk.edu.hk/policy/academichonesty/

Student/Faculty Expectations on Teaching and Learning

- https://www.cse.cuhk.edu.hk/~cslui/student_teacher_expectations.pdf

Use of ChatGPT, DeepSeek, Gemini (generative AI tools): Approach 3

- By default, generative AI tools are not allowed in all courses in the CUHK
- For this course, we release this constraints, you can use ChatGPT, Gemini, DeepSeek for the whole course (except the final exam)
- When you use the tools, write clearly which part, and how much you have used the tools in you work

Expectations from this course

1. Have basic understanding or **willing to learn** the Python **Programming** Language
2. Would like to **challenge** yourself **with** interesting programming and problems related to **different types of data**

Approach of this course

- Machine learning is a huge topic, to be good at machine learning, you will have to have good foundations in
  - Statistics
  - Linear algebra
  - Calculus
  - Probability theory
  - Signal processing
  - ...
- However, this is NOT just a course that follow the usual path of introduction to machine learning

Approach of this course

- In this course, we also focus on the following topics:
  - Data collection and preprocessing
  - Data representation
  - Data visualization
  - Problem specific domain
- What you will **learn** after taking this course?
  - Working on common machine learning tasks
  - Paying attention to the details to different problem domains
  - Visualizing the data at different stages
  - The intuition of neural representation of the data

Have your heard these terms before? What are they?

1. Data science
2. Data analysis
3. Big data
4. Database
5. Data mining
6. Machine learning
7. Deep learning
8. Artificial intelligence
9. Data scientist
10. Machine learning engineer
11. Data engineer
12. … etc

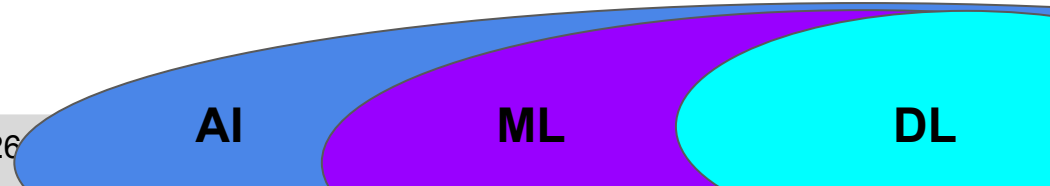
Terms related to “Data”

1. Database
   - A collection of data that is well-organized and easily managed
2. Data analysis
   - A process of retrieving knowledge from data
3. Data mining
   - A process of extracting knowledge in large data sets
4. Big data
   - A field that handle large and complex data sets
   - Challenges: data storage, data analysis, transferring data, searching data, visualizing data, updating data, … etc

Artificial Intelligence, Machine Learning, Deep Learning

- These are commonly used in data analysis

1. Artificial intelligence
   - Knowledge or behaviour demonstrated by machines
2. Machine learning
   - The study of computer algorithms that improve knowledge retrieval from data
   - A part of artificial intelligence
3. Deep learning
   - The use of multiple layers in the network to improve knowledge retrieval from data
   - A part of machine learning



Data Scientist, ML Engineer, Data Engineer

- These are roles, jobs

1. Data scientist transforms data to valuable insights and strategic decisions.
2. Machine learning engineer turns data-driven models into practical applications.
3. Data engineer works on the the foundation of a robust data infrastructure.

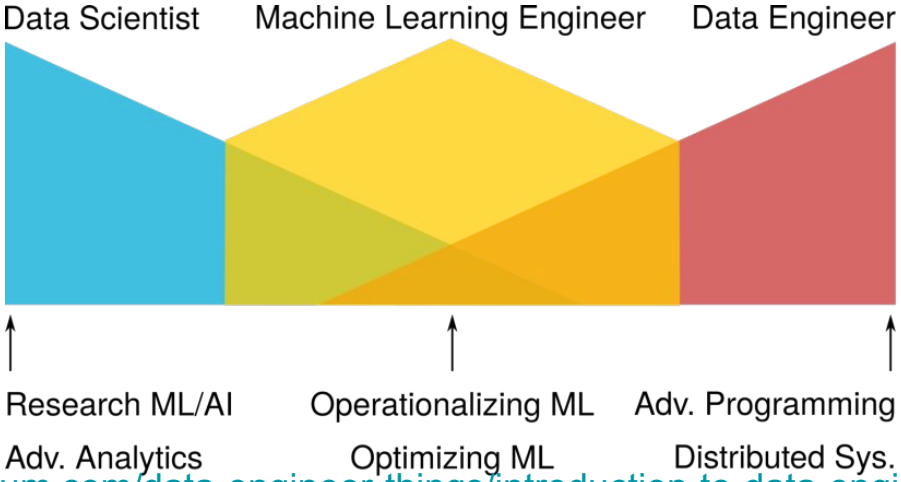


Image credit: https://medium.com/data-engineer-things/introduction-to-data-engineering-e16c9942dc2c

Data Science

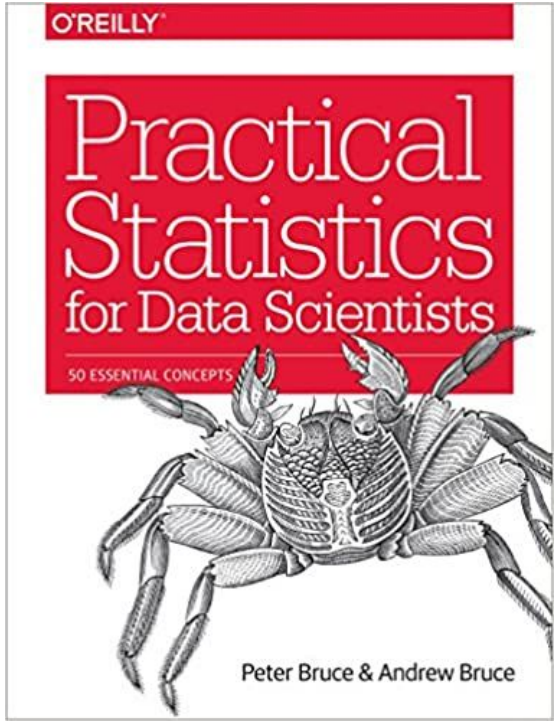
- It uses scientific methods, process, algorithms and systems to extract knowledge and insights from data
- It applies knowledge and actionable insights from data across a broad range of application domains
- Relevant disciplines include:
  - Computer science - database, data processing, programming
  - Informatics - study of computation systems, knowledge extraction
  - Statistics - interpretation of the results
  - Other disciplines - depend on the problem definitions
  - Communication and visualization

Reference Books

Practical Statistics for Data Scientists: 50 Essential Concepts

Chapters:

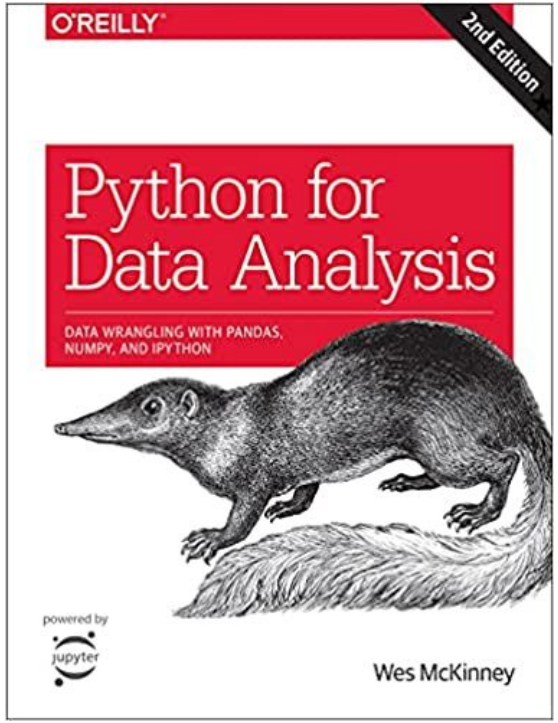
1. Exploratory Data Analysis
2. Data and Sampling Distributions
3. Statistical Experiments and Significance Testing
4. Regression and Prediction
5. Classification
6. Statistical Machine Learning
7. Unsupervised Learning



Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython

Chapters

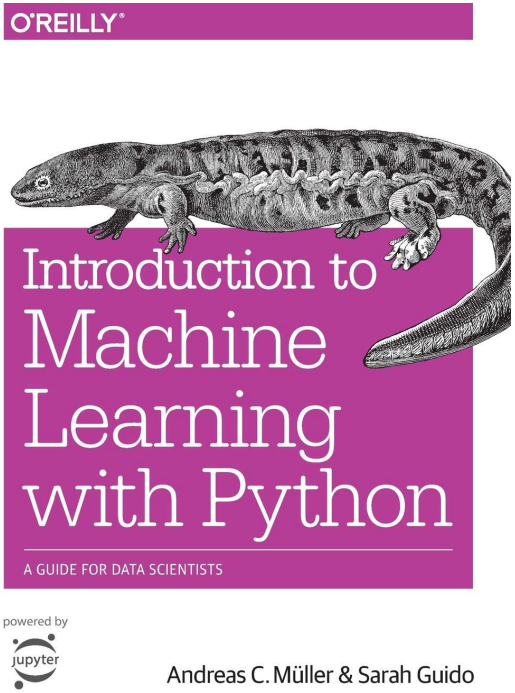
1. Preliminaries
2. Python language basics, IPython, and Jupyter notebooks
3. Built-in data structures, functions, and files
4. **NumPy basics: arrays and vectorized computation**
5. **Getting started with pandas**
6. **Data loading, storage, and file formats**
7. **Data cleaning and preparation**
8. **Data wrangling: join, combine, and reshape**
9. **Plotting and visualization**
10. **Data aggregation and group operations**
11. Time series
12. Advanced pandas
13. Introduction to modeling libraries in Python
14. Data analysis examples
15. Advanced NumPy
16. More on the IPython system



Introduction to Machine Learning with Python: A Guide for Data Scientists

Chapters:

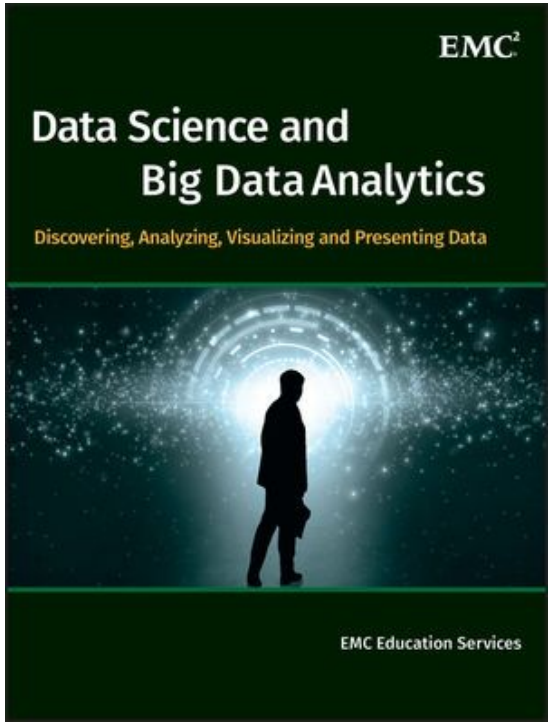
1. Introduction
2. Supervised Learning
3. Unsupervised Learning and Preprocessing
4. Representing Data and Engineering Features
5. Model Evaluation and Improvement
6. Algorithm Chains and Pipelines
7. Working with Text Data
8. Wrapping Up



Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data

Chapters:

1. Introduction to Big Data Analytics
2. Data Analytics Lifecycle
3. Review of Basic Data Analytic Methods Using R
4. Advanced Analytical Theory and Methods: Clustering
5. Advanced Analytical Theory and Methods: Association Rules
6. Advanced Analytical Theory and Methods: Regression
7. Advanced Analytical Theory and Methods: Classification
8. Advanced Analytical Theory and Methods: Time Series Analysis
9. Advanced Analytical Theory and Methods: Text Analysis
10. Advanced Analytics - Technology and Tools: MapReduce and Hadoop
11. Advanced Analytics - Technology and Tools: In-Database Analytics
12. The Endgame, or Putting It All Together



Deep Learning

Chapters:

1. Introduction

Part I: Applied Math and Machine Learning Basics

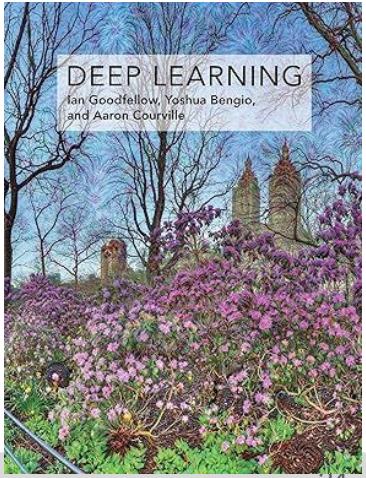
2. Linear Algebra
3. Probability and Information Theory
4. Numerical Computation
5. Machine Learning Basics

Part II: Modern Practical Deep Networks

6. Deep Feedforward Networks
7. Regularization for Deep Learning
8. Optimization for Training Deep Models
9. Convolutional Networks
10. Sequence Modeling: Recurrent and Recursive Nets
11. Practical Methodology
12. Applications

Part III: Deep Learning Research

13. Linear Factor Models
14. Autoencoders
15. Representation Learning
16. Structured Probabilistic Models for Deep Learning
17. Monte Carlo Methods
18. Confronting the Partition Function
19. Approximate Inference
20. Deep Generative Models



Reference: https://www.deeplearningbook.org/